Ramdeobaba University, Nagpur-13.

Department of Electronics Engineering

Embedded System Design Lab

Even Sem Semester – 2025-26

Experiment No - 4

Aim: **Write an Assembly language program to arrange 10 number in ascending order present in memory and store the result into memory**

Name of Student:	Anshika Yadav
Batch / Roll No.	B2/ 21
Semester/Section:	4th/ A
Date of Performance:	5/2/26
Date of Submission:	
Name & Signature of Faculty	___Dr. Vicky Butram_______________________________________ ________

Experiment No-04

Aim: Write an Assembly Language Program to check whether a number stored in memory is odd or even.

Software Used: Keil uVision5

Program:

```
        AREA ODD_EVEN, CODE, READONLY
        EXPORT __main

__main
        LDR     R0, =0x1FFFF000
        LDR     R1, [R0]

        ANDS    R2, R1, #1
        CMP     R2, #0
        BEQ     EVEN

ODD     MOVS    R3, #1
        B       STORE

EVEN    MOVS    R3, #0

STORE   LDR     R4, =0x1FFFF004
        STR     R3, [R4]

STOP    B       STOP
```

Program Screenshot

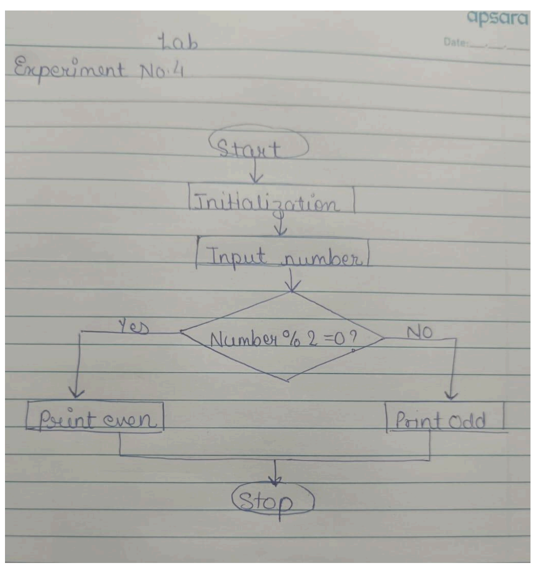
```
lab4.s   lab4.s   system_MKL25Z4.c

            AREA ODD_EVEN, CODE, READONLLY
            EXPORT __main

__main
            LDR R0, =0X1FFFF000
            LDR R1, [R0]
ANDS        R2, R1, #1
            CMP R2,#0
            BEQ EVEN
ODD         MOVS R3, #1
            B STORE
EVEN        MOVS R3, #0
            STORE
STORE       LDR R4, =0X1FFF004
            STR R3,[R4]
STOP        B STOP

            LDR R4, =0X1FFFF004
            STR R3,[R4]
END         B STOP
```

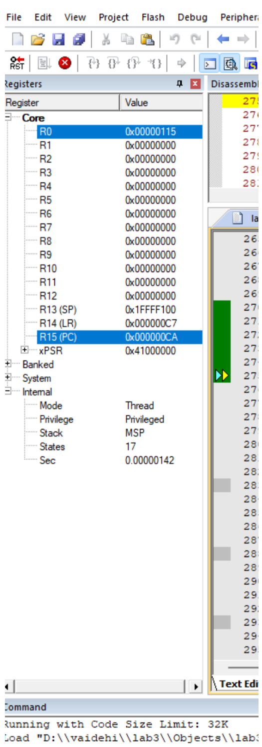
Flowchart:



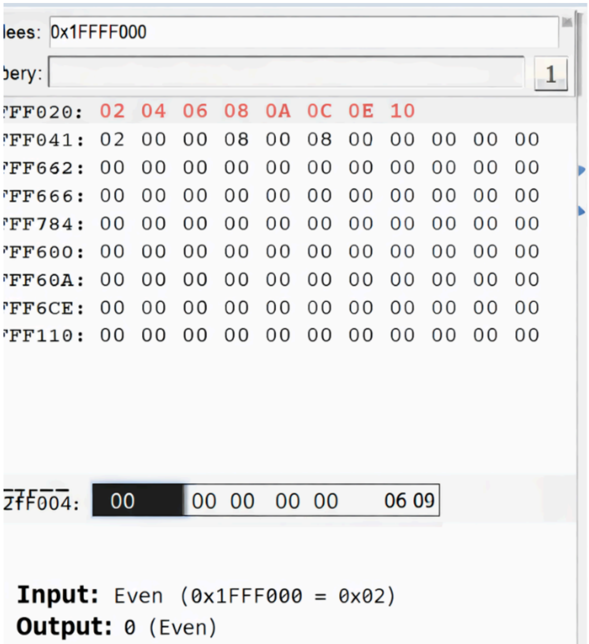
Results and Conclusion:

The ARM assembly program to determine whether a number is odd or even was executed successfully. The program loads a 32-bit number from memory location 0x1FFFF000 into register R1 and performs a bitwise AND operation with 1 using the instruction ANDS R2, R1, #1 to check the Least Significant Bit (LSB). Since the LSB determines whether a number is odd or even, a result of 0 indicates an even number and a result of 1 indicates an odd number. Based on the comparison, the program branches accordingly and stores 0 (for even) or 1 (for odd) at memory location 0x1FFFF004 using the STR instruction. The output observed in memory confirms that the correct value is stored depending on the input number. Thus, the objective of determining whether a number is odd or even using ARM assembly language and bitwise operations was successfully achieved, demonstrating the use of conditional branching, logical instructions, and memory handling in ARM programming.

Output:



Memory



Numbers were identified as even or odd and stored in memory